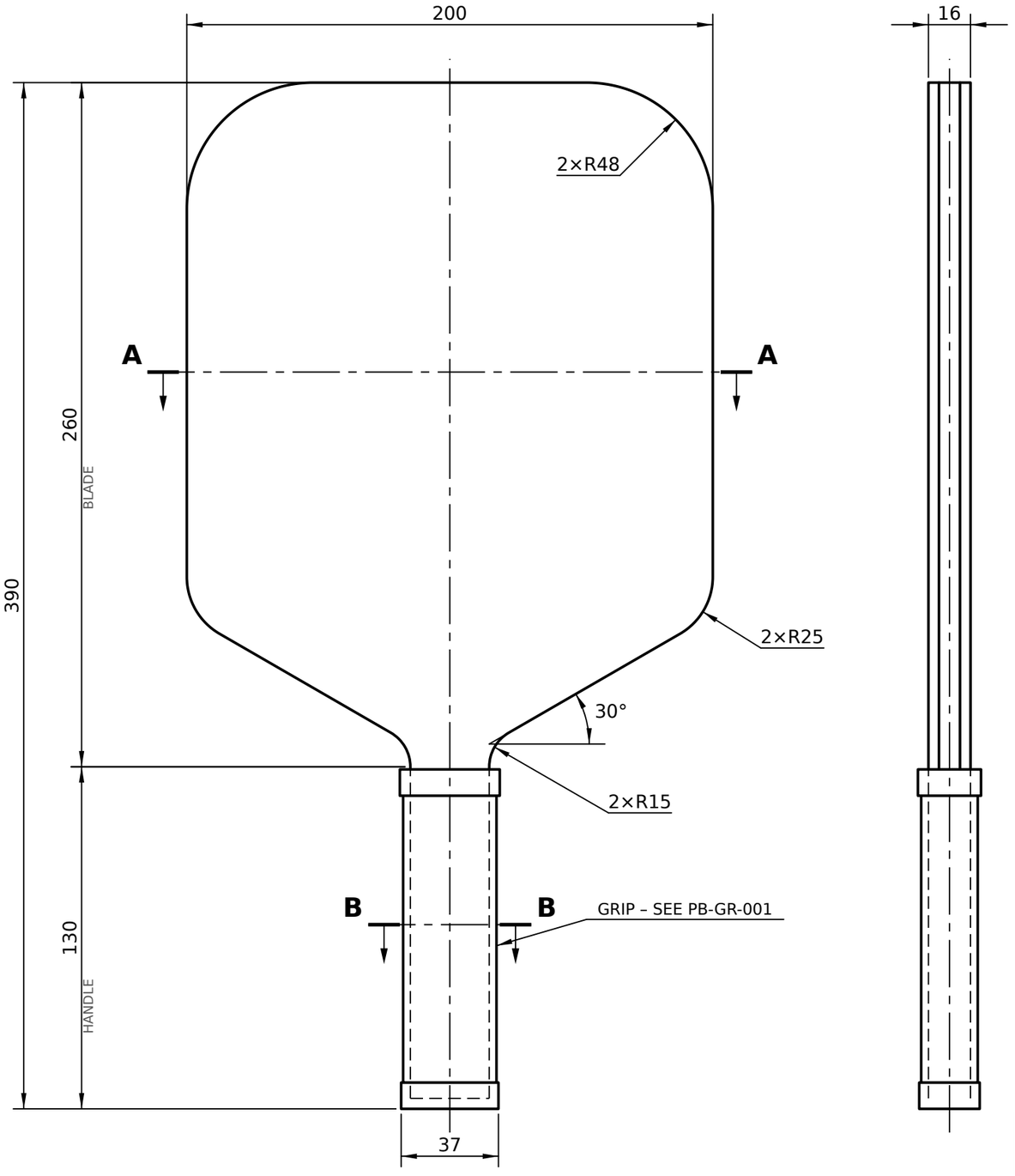
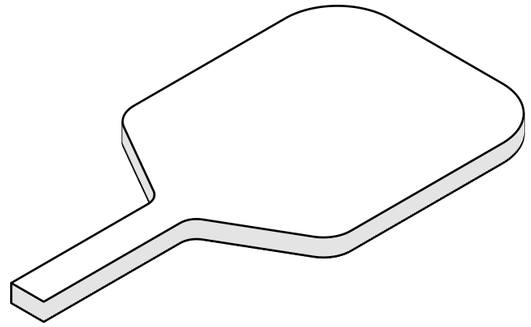


FRONT VIEW

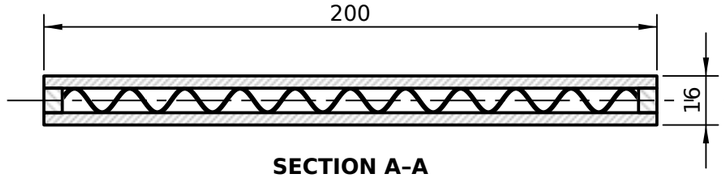
SIDE VIEW

LAY-UP 4 / 8 / 4



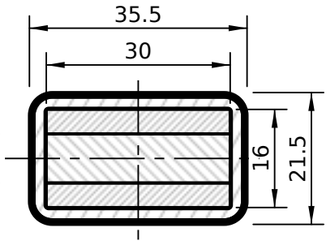
ISOMETRIC VIEW

SCALE 1:4 (BODY ONLY)



SECTION A–A

SCALE 1:2 · CF 4 / WAVE-RIB CORE 8 / CF 4 · CORE DETAIL: SHEET 3



SECTION B–B

SCALE 1:1 · GRIP: PB-GR-001

NOTES

1. ALL DIMENSIONS IN MILLIMETRES. DO NOT SCALE THE DRAWING. UNLESS STATED: LINEAR ±0.5, ANGULAR ±0.5°.
2. PROFILE SYMMETRICAL ABOUT THE CENTRE LINE. ALL ARCS TANGENT TO ADJOINING EDGES.
3. SANDWICH: CF FACE SHEET 4 + WAVE-RIB CORE 8 + CF FACE SHEET 4 = 16 ±0.3, UNIFORM OVER BLADE AND HANDLE. BODY (SHEETS + CORE) FROM y = 4 TO THE TIP; BUTT CAP y = 0 … 10.
4. FACE SHEETS (2 OFF, IDENTICAL): WOVEN CARBON/EPOXY PLATE 4 THK, CUT TO PB-EL-001_profile.dxf, OUTER EDGES R0.8. TEMPLATE: SHEET 2.
5. CORE: 3D-PRINTED PA-CF, CLOSED FRAME 6, WAVE RIB λ18 / WALL 1.6 / LANDS 4.25, SOLID TO y = 150 WITH 45° FLUTE RUN-OUT – SEE SHEET 3.
6. BOND CORE TO SHEETS: STRUCTURAL EPOXY, ABRADED + PRIMED SURFACES, FILLET ALL LAND EDGES (CRITICAL ITEM – SEE THE STRESS REPORT).
7. GRIP: SLEEVE + WRAP + COLLAR + BUTT CAP, ISO 7380 M4×12 SCREW INTO M4 HEAT-SET INSERT – SEE PB-GR-001. GRIP 35.5 × 21.5, CIRCUMFERENCE 107.6 (4¼ in).
8. USA PICKLEBALL LIMITS: L + W = 590 ≤ 609.6 (24 in), L = 390 ≤ 431.8 (17 in), INCL. BUTT CAP. AN EDGE GUARD COUNTS TOWARDS THESE LIMITS (MARGIN 19.6).
9. MASS 832 g, BALANCE POINT 238 FROM THE BUTT (FROM THE 3D MODEL).

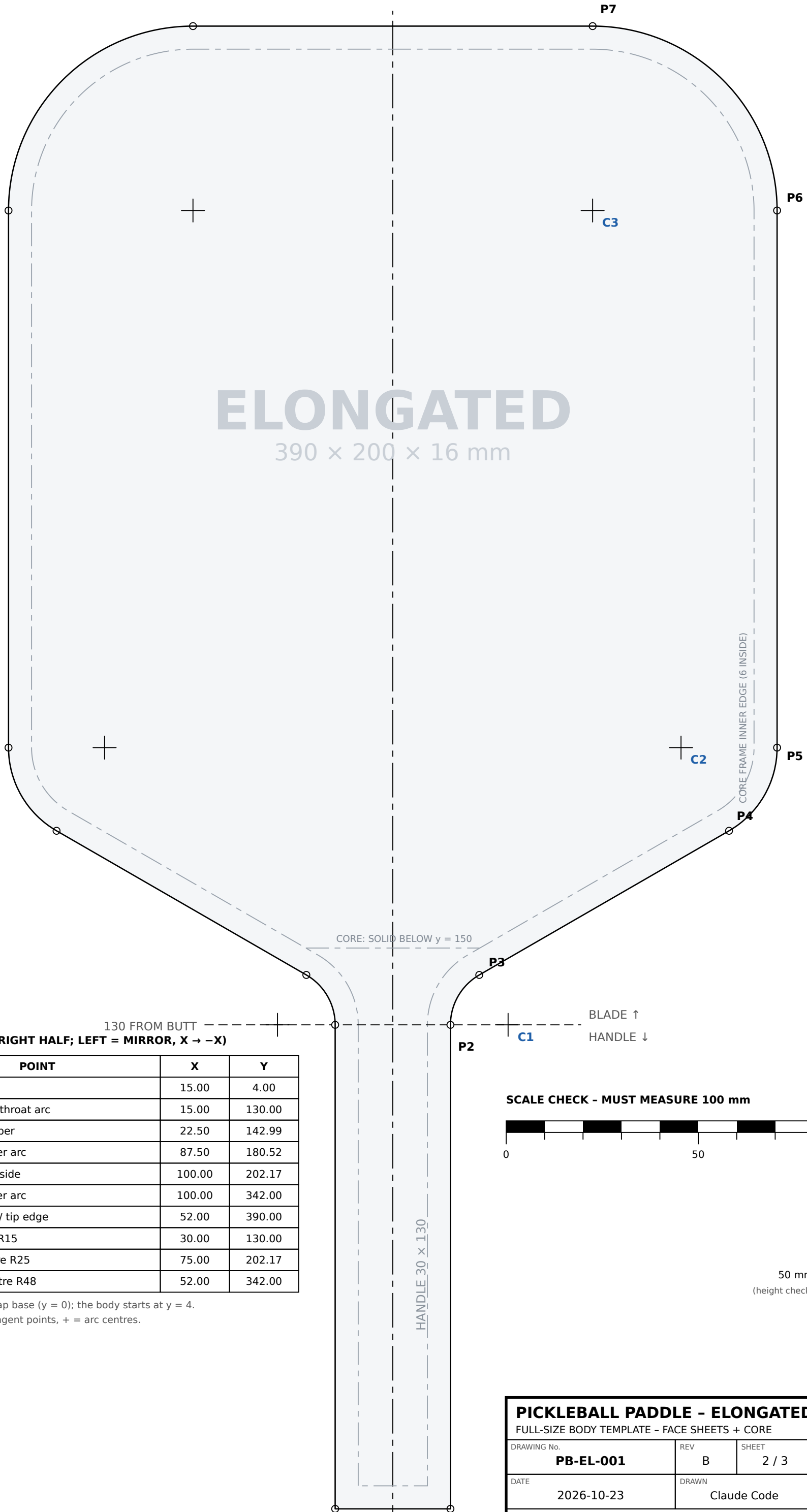
PICKLEBALL PADDLE – ELONGATED

GENERAL ARRANGEMENT · 390 × 200 × 16 mm · CF / WAVE-RIB SANDWICH

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-23		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-EL-001	B	1 / 3	A3	1:2	mm

PROJECTION	MATERIAL	STATUS
	SEE NOTES	ISSUED FOR REVIEW

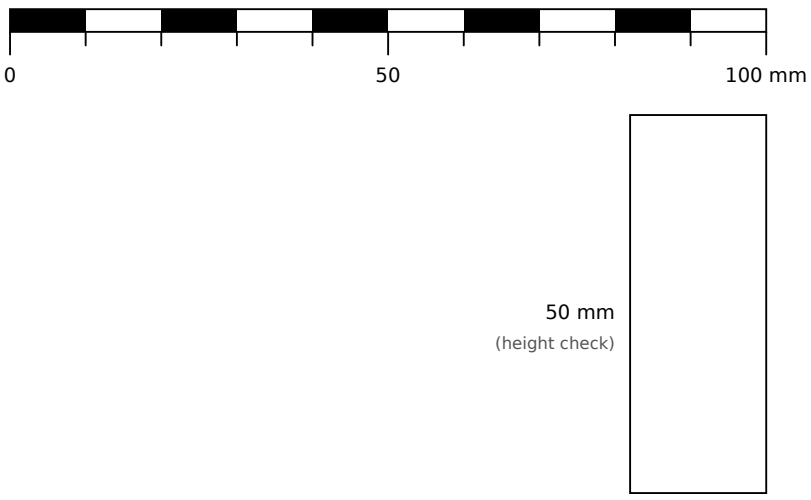


PROFILE GEOMETRY (RIGHT HALF; LEFT = MIRROR, X → −X)

ID	POINT	X	Y
P1	Butt corner	15.00	4.00
P2	Handle edge / throat arc	15.00	130.00
P3	Throat arc / taper	22.50	142.99
P4	Taper / shoulder arc	87.50	180.52
P5	Shoulder arc / side	100.00	202.17
P6	Side / tip-corner arc	100.00	342.00
P7	Tip-corner arc / tip edge	52.00	390.00
C1	Throat centre R15	30.00	130.00
C2	Shoulder centre R25	75.00	202.17
C3	Tip-corner centre R48	52.00	342.00

Origin: centre of the butt-cap base (y = 0); the body starts at y = 4.
All arcs tangent. Dots = tangent points, + = arc centres.

SCALE CHECK – MUST MEASURE 100 mm



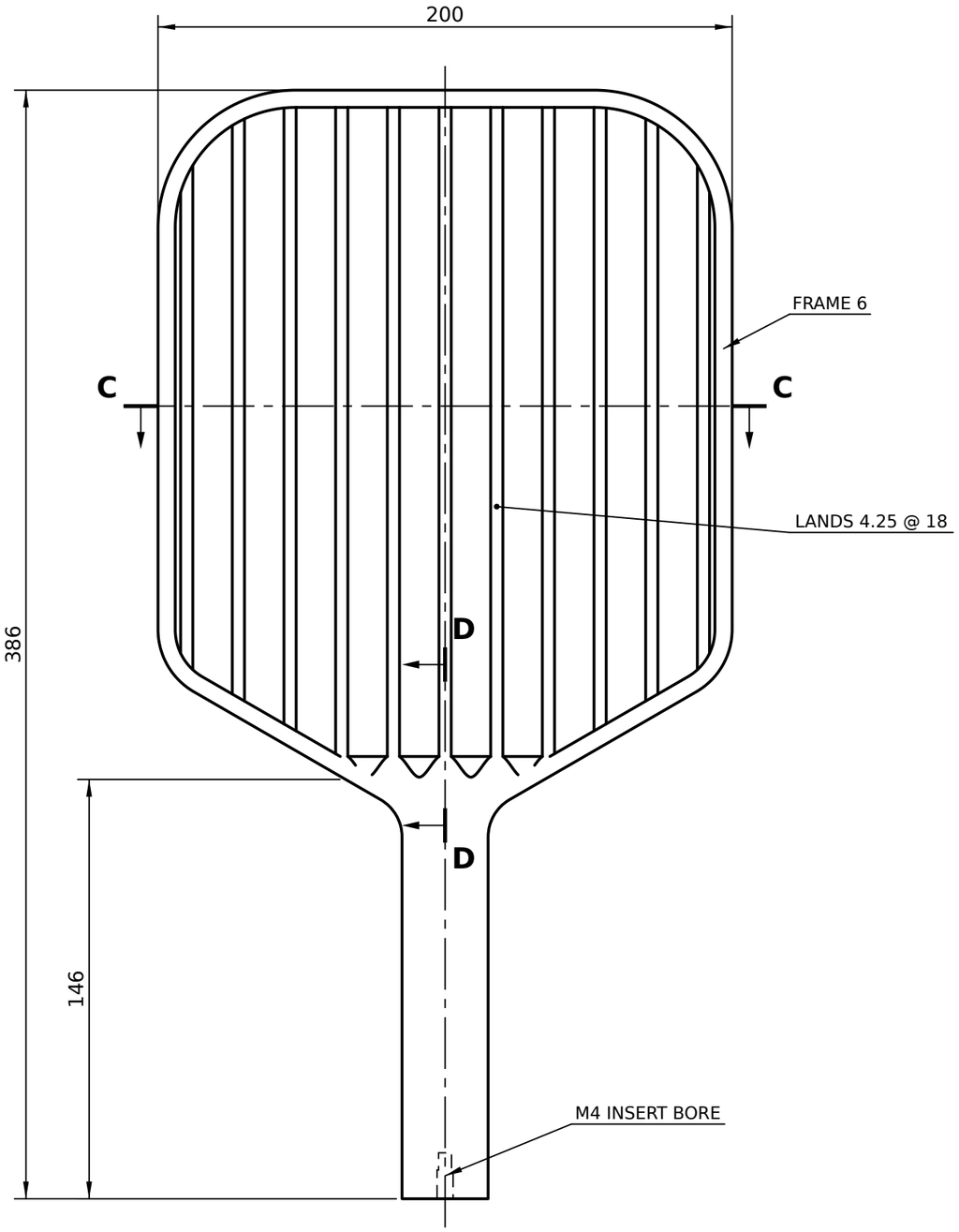
PICKLEBALL PADDLE – ELONGATED

FULL-SIZE BODY TEMPLATE – FACE SHEETS + CORE

DRAWING No.	REV	SHEET	SCALE
PB-EL-001	B	2 / 3	**1:1**

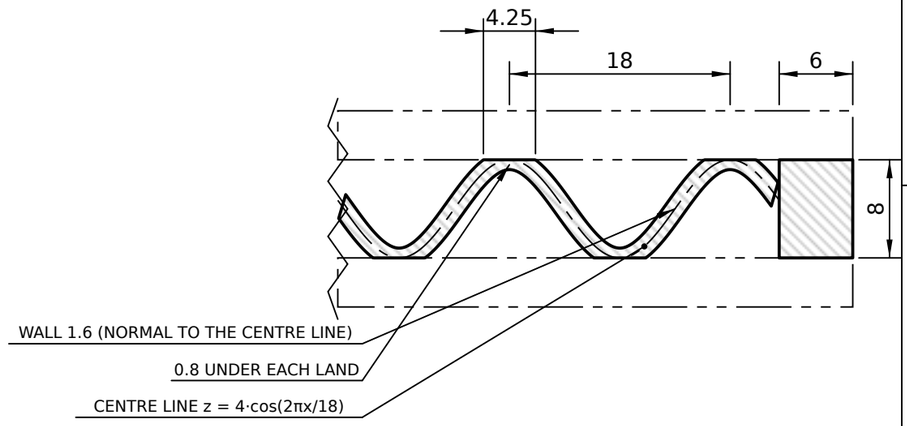
DATE	DRAWN	UNITS
2026-10-23	Claude Code	mm

PRINT ON A3 AT 100 % / "ACTUAL SIZE". NO FIT-TO-PAGE.
Verify both scale bars before cutting.



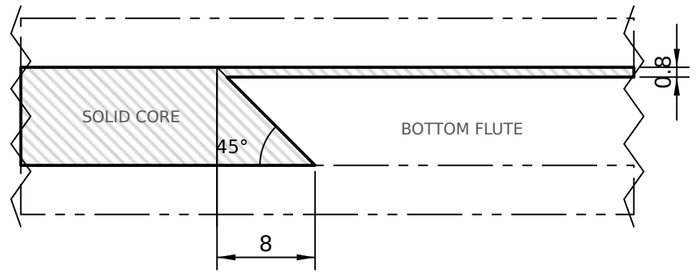
PLAN VIEW (TOP FACE)

SCALE 1:2 · BOTTOM LANDS OFFSET ½ PITCH



SECTION C–C (PART)

SCALE 2:1 · FACE SHEETS SHOWN PHANTOM



SECTION D–D (PART)

SCALE 2:1 · ON THE CENTRE LAND (x = 0), RUN-OUT FROM y = 150; TOP FLUTES MIRROR IT

NOTES

1. MATERIAL: PA-CF (CARBON-FIBRE-FILLED NYLON), E ≈ 4 GPa, ρ 1.18 g/cm³. DRY THE FILAMENT BEFORE PRINTING.
2. PRINT FLAT, 100 % SOLID: WALL 1.6 = 4 PERIMETERS (0.4 NOZZLE), 0.2 LAYERS. NO SUPPORTS (45° RUN-OUTS).
3. THICKNESS 8 ±0.1. BOTH FACES ARE BOND SURFACES, FLAT WITHIN 0.1.
4. WAVE RIB λ18, WALL 1.6, CENTRE-LINE AMPLITUDE 4; CRESTS CUT FLAT = LANDS 4.25 ON BOTH FACES.
5. FRAME 6, CLOSED. SOLID TO y = 150; EACH FLUTE STARTS WITH A 45° RUN-OUT, 8 LONG (D–D).
6. M4 INSERT BORE Ø5.6 × 10 + Ø4.5 × 16 IN THE BUTT FACE – SEE PB-GR-001.
7. BONDING: ABRADE (P120), DEGREASE, STRUCTURAL EPOXY, FILLET EVERY LAND EDGE.
8. MASS 179 g. 3D MODEL: output/cad/elongated_rib_core.step / .stl. y FROM THE BUTT-CAP BASE (CORE STARTS AT y = 4).

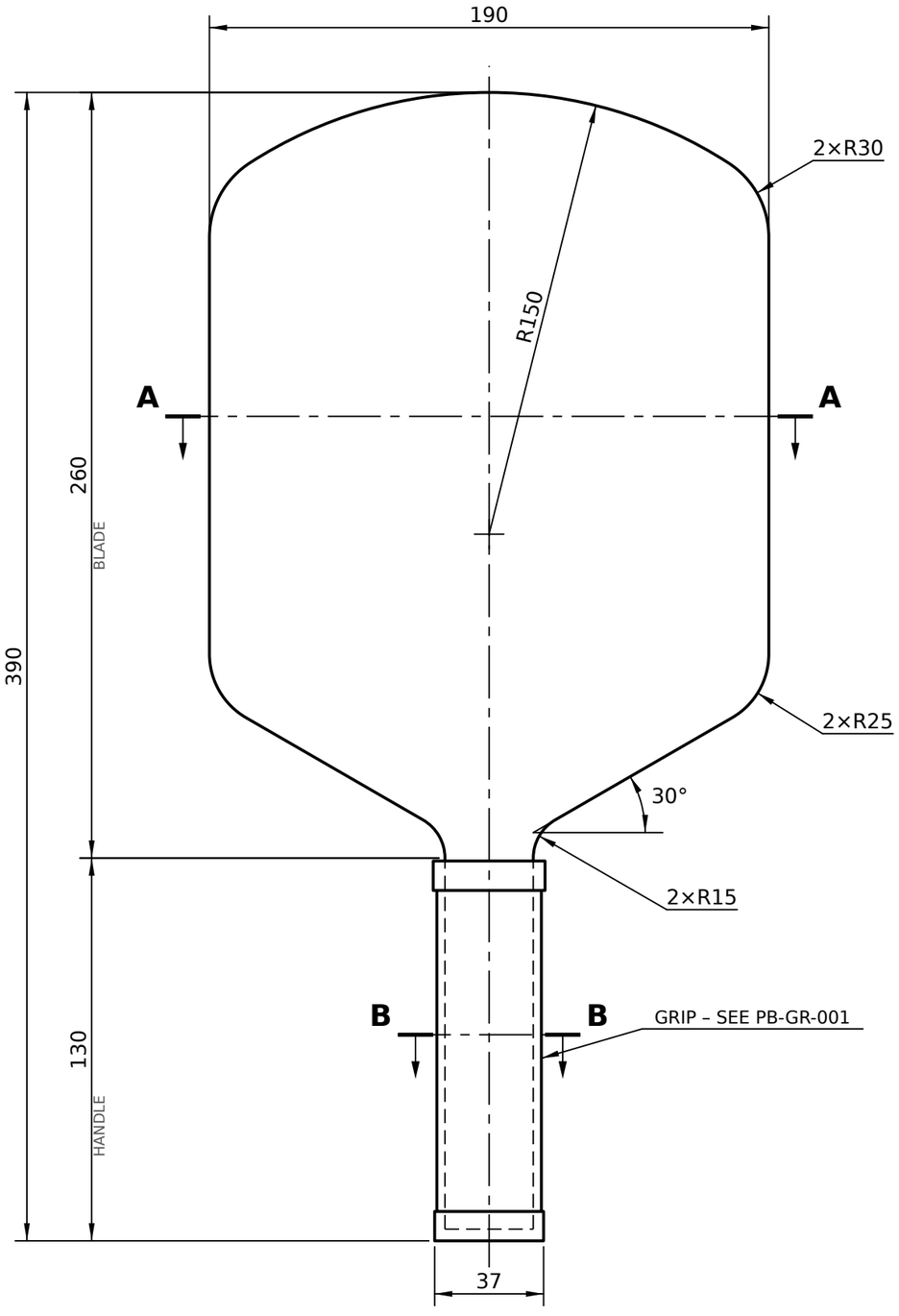
RIB CORE – ELONGATED

RIB CORE · 3D-PRINTED PA-CF · 386 × 200 × 8 mm

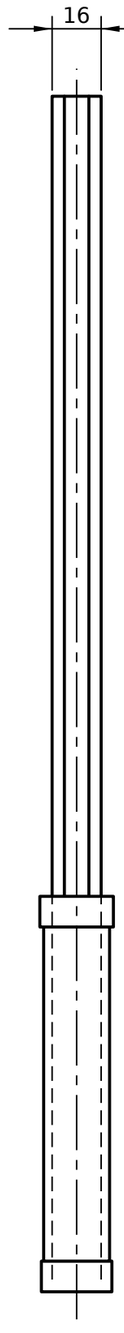
DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-23		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-EL-001	B	3 / 3	A3	1:2, 2:1	mm

PROJECTION	MATERIAL	STATUS
	PA-CF (NOTE 1)	ISSUED FOR REVIEW

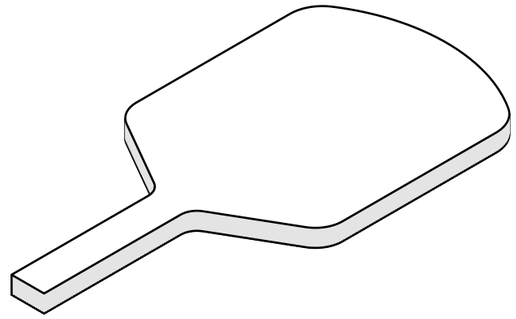


FRONT VIEW



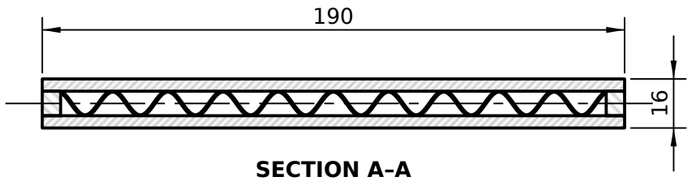
SIDE VIEW

LAY-UP 4 / 8 / 4



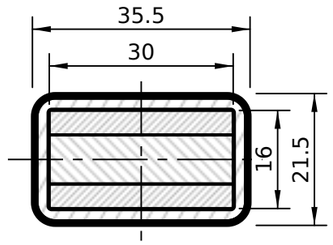
ISOMETRIC VIEW

SCALE 1:4 (BODY ONLY)



SECTION A–A

SCALE 1:2 · CF 4 / WAVE-RIB CORE 8 / CF 4 · CORE DETAIL: SHEET 3



SECTION B–B

SCALE 1:1 · GRIP: PB-GR-001

NOTES

1. ALL DIMENSIONS IN MILLIMETRES. DO NOT SCALE THE DRAWING. UNLESS STATED: LINEAR ±0.5, ANGULAR ±0.5°.
2. PROFILE SYMMETRICAL ABOUT THE CENTRE LINE. ALL ARCS TANGENT TO ADJOINING EDGES.
3. SANDWICH: CF FACE SHEET 4 + WAVE-RIB CORE 8 + CF FACE SHEET 4 = 16 ±0.3, UNIFORM OVER BLADE AND HANDLE. BODY (SHEETS + CORE) FROM y = 4 TO THE TIP; BUTT CAP y = 0 … 10.
4. FACE SHEETS (2 OFF, IDENTICAL): WOVEN CARBON/EPOXY PLATE 4 THK, CUT TO PB-HY-001_profile.dxf, OUTER EDGES R0.8. TEMPLATE: SHEET 2.
5. CORE: 3D-PRINTED PA-CF, CLOSED FRAME 6, WAVE RIB λ18 / WALL 1.6 / LANDS 4.25, SOLID TO y = 150 WITH 45° FLUTE RUN-OUT – SEE SHEET 3.
6. BOND CORE TO SHEETS: STRUCTURAL EPOXY, ABRADED + PRIMED SURFACES, FILLET ALL LAND EDGES (CRITICAL ITEM – SEE THE STRESS REPORT).
7. GRIP: SLEEVE + WRAP + COLLAR + BUTT CAP, ISO 7380 M4×12 SCREW INTO M4 HEAT-SET INSERT – SEE PB-GR-001. GRIP 35.5 × 21.5, CIRCUMFERENCE 107.6 (4¼ in).
8. USA PICKLEBALL LIMITS: L + W = 580 ≤ 609.6 (24 in), L = 390 ≤ 431.8 (17 in), INCL. BUTT CAP. AN EDGE GUARD COUNTS TOWARDS THESE LIMITS (MARGIN 29.6).
9. MASS 785 g, BALANCE POINT 233 FROM THE BUTT (FROM THE 3D MODEL).

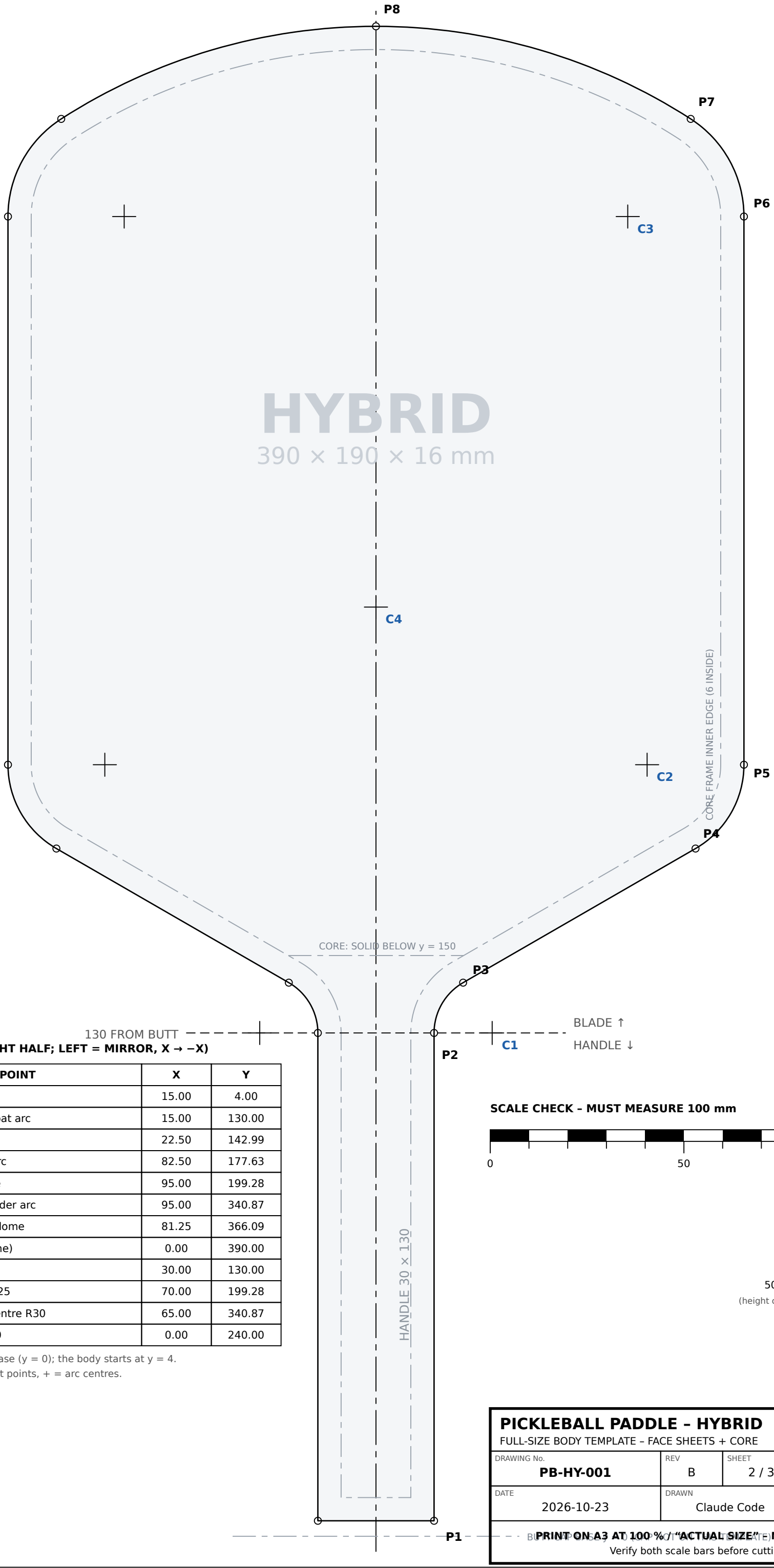
PICKLEBALL PADDLE – HYBRID

GENERAL ARRANGEMENT · 390 × 190 × 16 mm · CF / WAVE-RIB SANDWICH

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-23		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	B	1 / 3	A3	1:2	mm

PROJECTION	MATERIAL	STATUS
	SEE NOTES	ISSUED FOR REVIEW



PROFILE GEOMETRY (RIGHT HALF; LEFT = MIRROR, X → −X)

ID	POINT	X	Y
P1	Butt corner	15.00	4.00
P2	Handle edge / throat arc	15.00	130.00
P3	Throat arc / taper	22.50	142.99
P4	Taper / shoulder arc	82.50	177.63
P5	Shoulder arc / side	95.00	199.28
P6	Side / crown-shoulder arc	95.00	340.87
P7	Crown shoulder / dome	81.25	366.09
P8	Apex (on centre line)	0.00	390.00
C1	Throat centre R15	30.00	130.00
C2	Shoulder centre R25	70.00	199.28
C3	Crown-shoulder centre R30	65.00	340.87
C4	Dome centre R150	0.00	240.00

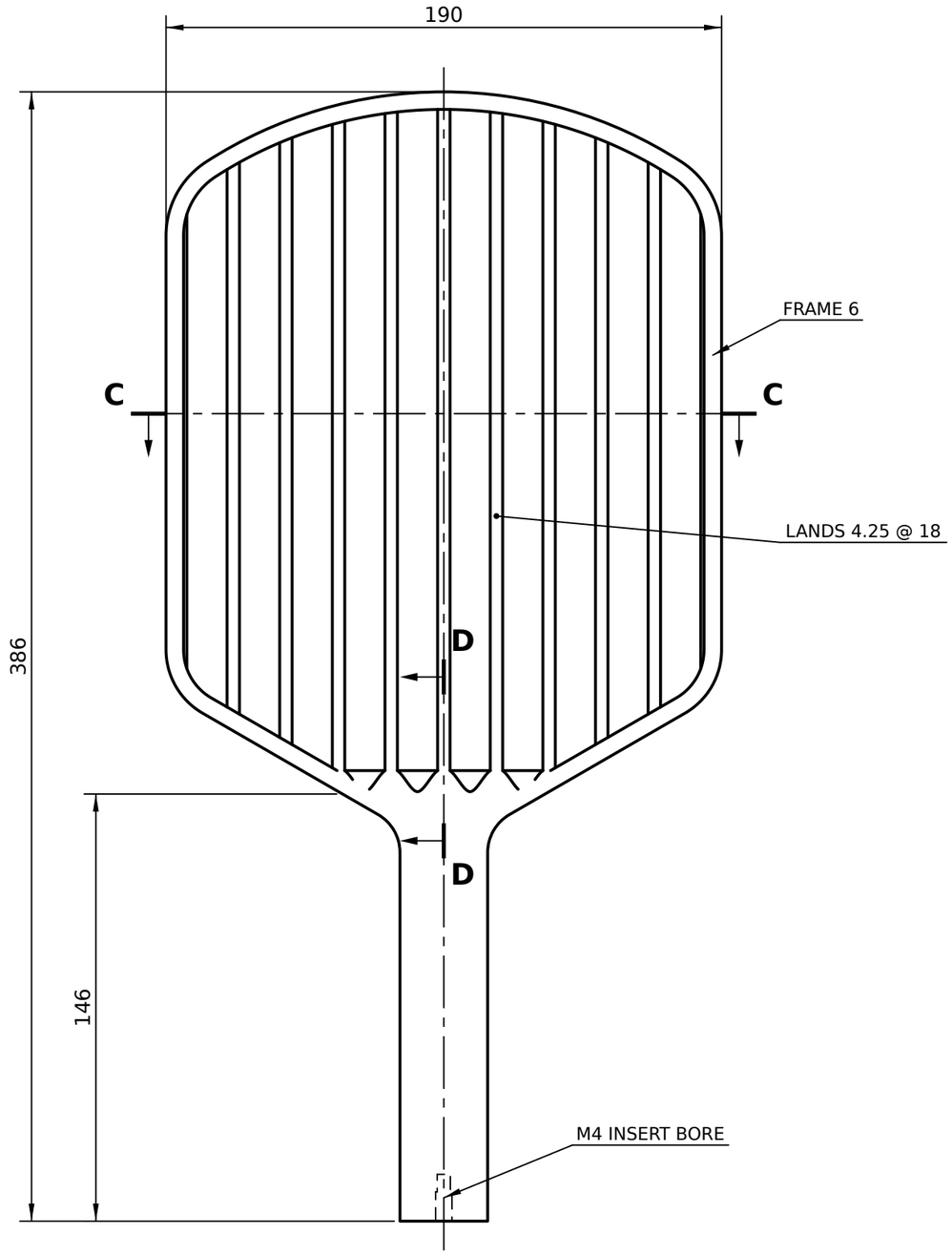
Origin: centre of the butt-cap base (y = 0); the body starts at y = 4.
All arcs tangent. Dots = tangent points, + = arc centres.

PICKLEBALL PADDLE – HYBRID

FULL-SIZE BODY TEMPLATE – FACE SHEETS + CORE

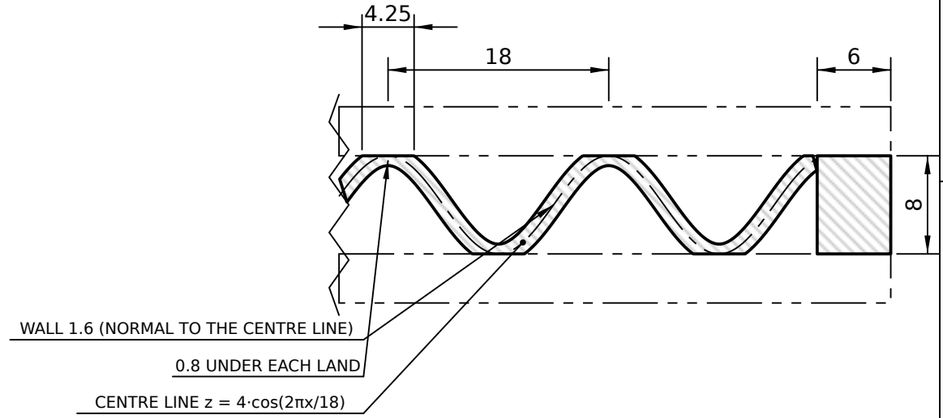
DRAWING No.	REV	SHEET	SCALE
PB-HY-001	B	2 / 3	**1:1**
DATE	DRAWN	UNITS	
2026-10-23	Claude Code	mm	

PRINT ON A3 AT 100 % / "ACTUAL SIZE". NO FIT-TO-PAGE.
Verify both scale bars before cutting.



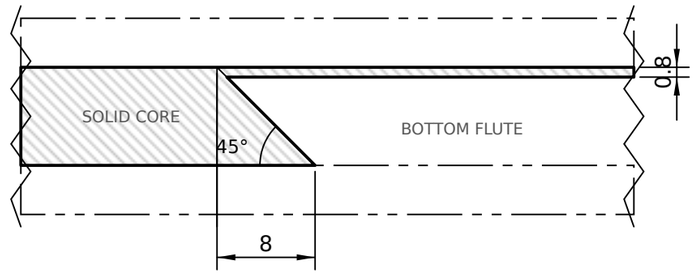
PLAN VIEW (TOP FACE)

SCALE 1:2 · BOTTOM LANDS OFFSET ½ PITCH



SECTION C–C (PART)

SCALE 2:1 · FACE SHEETS SHOWN PHANTOM



SECTION D–D (PART)

SCALE 2:1 · ON THE CENTRE LAND (x = 0), RUN-OUT FROM y = 150; TOP FLUTES MIRROR IT

NOTES

1. MATERIAL: PA-CF (CARBON-FIBRE-FILLED NYLON), E ≈ 4 GPa, ρ 1.18 g/cm³. DRY THE FILAMENT BEFORE PRINTING.
2. PRINT FLAT, 100 % SOLID: WALL 1.6 = 4 PERIMETERS (0.4 NOZZLE), 0.2 LAYERS. NO SUPPORTS (45° RUN-OUTS).
3. THICKNESS 8 ±0.1. BOTH FACES ARE BOND SURFACES, FLAT WITHIN 0.1.
4. WAVE RIB λ18, WALL 1.6, CENTRE-LINE AMPLITUDE 4; CRESTS CUT FLAT = LANDS 4.25 ON BOTH FACES.
5. FRAME 6, CLOSED. SOLID TO y = 150; EACH FLUTE STARTS WITH A 45° RUN-OUT, 8 LONG (D–D).
6. M4 INSERT BORE Ø5.6 × 10 + Ø4.5 × 16 IN THE BUTT FACE – SEE PB-GR-001.
7. BONDING: ABRADE (P120), DEGREASE, STRUCTURAL EPOXY, FILLET EVERY LAND EDGE.
8. MASS 170 g. 3D MODEL: output/cad/hybrid_rib_core.step / .stl. y FROM THE BUTT-CAP BASE (CORE STARTS AT y = 4).

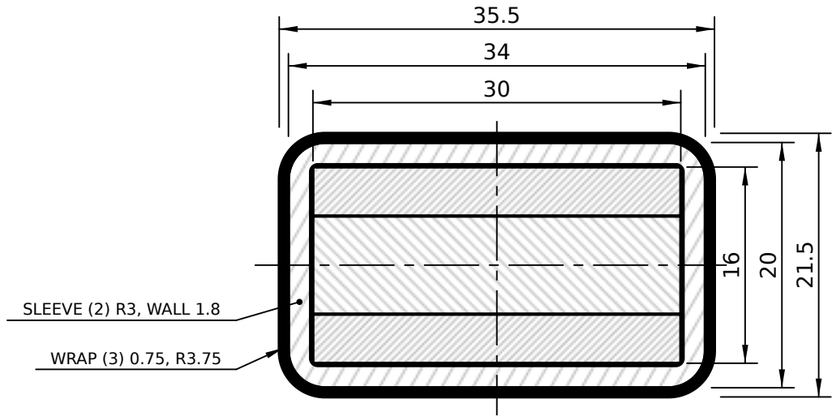
RIB CORE – HYBRID

RIB CORE · 3D-PRINTED PA-CF · 386 × 190 × 8 mm

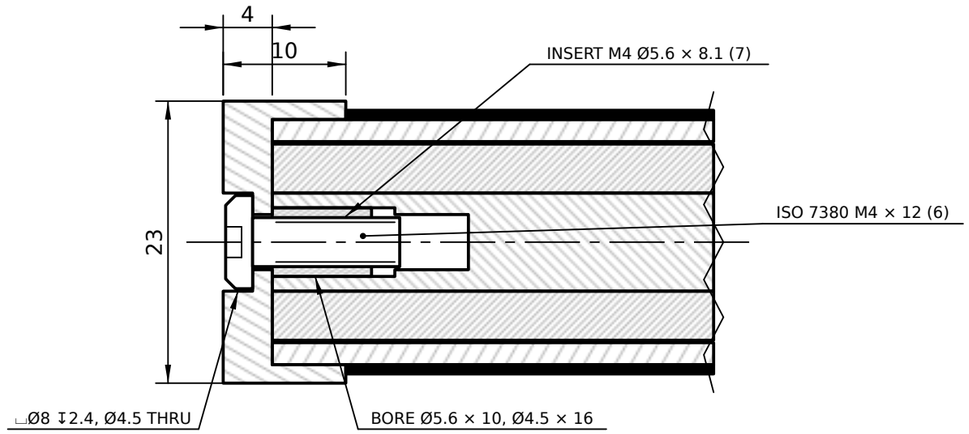
DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-23		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	B	3 / 3	A3	1:2, 2:1	mm

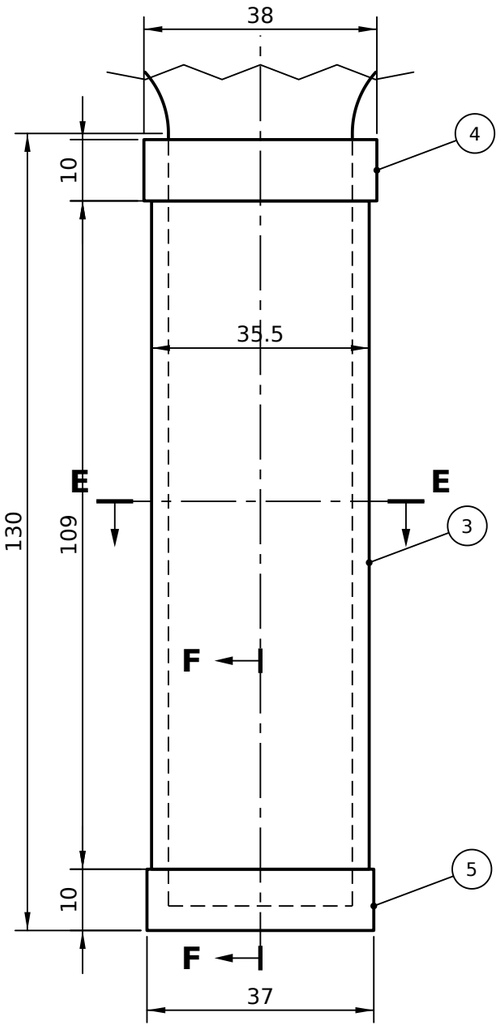
PROJECTION	MATERIAL	STATUS
	PA-CF (NOTE 1)	ISSUED FOR REVIEW



SECTION E–E
SCALE 2:1



SECTION F–F (BUTT END)
SCALE 2:1



HANDLE (FACE VIEW)
SCALE 1:1

ITEM	PART	MATERIAL / SPECIFICATION	QTY	MASS g (EL / HY)
1	Face sheet	Woven carbon/epoxy plate 4 thk, cut to profile DXF	2	304 / 285
1a	Rib core	PA-CF, 3D printed – sheet 3 of PB-EL-001 / PB-HY-001	1	179 / 170
2	Grip sleeve	TPU 95A, printed or moulded, 34 × 20 R3	1	26
3	Grip wrap	PU replacement grip 0.75 thk, 109 long	1	9.5
4	Throat collar	TPU 95A, 38 × 24 × 10	1	1.7
5	Butt cap	TPU 95A, 37 × 23, base 4 + skirt 6	1	5.0
6	Screw	ISO 7380-1 M4 × 12 button head, A2 stainless	1	1.8
7	Threaded insert	M4 heat-set, brass, OD 5.6 × 8.1	1	1.1
8	Adhesive	Structural epoxy (core–sheets, sleeve, collar)	a/r	–
COMPLETE PADDLE (3D MODEL)				**832 / 785**

NOTES

1. GRIP: 35.5 × 21.5 R3.75 OVER THE WRAP = CIRCUMFERENCE 107.6 (4¼ in). WRAP LENGTH 109.
2. HANDLE 130 INCL. BUTT CAP. BODY 30 × 16 (CF 4 / CORE 8 / CF 4).
3. SLEEVE (2): BOND TO THE HANDLE WITH EPOXY, 0.2 GAP PER SIDE FOR THE ADHESIVE.
4. WRAP (3): REPLACEABLE PU GRIP, SPIRAL FROM THE CAP SKIRT UP TO UNDER THE COLLAR.
5. COLLAR (4): SLIDE OVER THE WRAP END, BOND TO THE SLEEVE; TOP AT y = 129.
6. INSERT (7): HEAT-SET AT ≈250 °C, FLUSH WITH THE CORE BUTT FACE (y = 4), SQUARE WITHIN 1°.
7. SCREW (6): LOW-STRENGTH THREADLOCKER; TIGHTEN UNTIL THE CAP SEATS, MAX 0.8 N·m.
8. BUTT CAP (5): BASE 4 + SKIRT 6 OVER THE SLEEVE; THE BASE COUNTS IN THE 390 OVERALL LENGTH.
9. SAME HANDLE ON PB-EL-001 AND PB-HY-001. ALL DIMENSIONS IN mm.

GRIP AND BUTT-CAP ASSEMBLY

HANDLE 130 × 30 × 16 · GRIP 107.6 (4¼ in) · BOTH PADDLES

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-23		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-GR-001	B	1 / 1	A3	1:1, 2:1	mm

PROJECTION	MATERIAL	STATUS
	SEE BOM	ISSUED FOR REVIEW